Lecture 17: Taylor and Maclaurin Series

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Understand and work out the series representation of a function.
- ☐ Define Taylor and Maclaurin series and identify the conditions under which they represent a function.
- ☐ Derive the Taylor and Maclaurin series for a given function using derivatives at a point.
- ☐ Compute Taylor polynomials of specified degrees to approximate functions near a point.
- ☐ Expand elementary functions using Maclaurin series.

Series Representation

In last lecture, we have seen how to generate a power series for certain function by using the geometric series. But now if we are given any function, can we always represent it with a power series?

Taylor series - a power series generated by a function that are infinitely differentiable.

Within the interval of convergence I, a power series is a continuous function with derivative of all orders. How about the other way around?

Assume that $f(x)$ is the sum of a power series about $x = a$,

$$\begin{aligned} f(x) &= \sum_{n=0}^{\infty} a_n (x-a)^n \\ &= a_0 + a_1(x-a) + a_2(x-a)^2 + \cdots + a_n(x-a)^n + \cdots \end{aligned}$$

with a positive radius of convergence.

Series Representation

Taylor and Maclaurin Series

Let f be a function with derivatives of all orders throughout some interval containing a as an interior point. Then the Taylor series generated by f at $x = a$ is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n = f(a) + f'(a)(x-a) + \frac{f''(x)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \cdots .$$

The Maclaurin series of f is the Taylor series generated by f at $x = 0$, or

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + \cdots .$$

Taylor and Maclaurin Series

Find the Taylor series generated by $f(x) = \frac{1}{x}$ at $a = 2$. Where, if anywhere, does the series converge to $\frac{1}{x}$?

Taylor Polynomials

Let f be a function with derivatives of order k for $k - 1, 2, \ldots, N$ in some interval containing a as an interior point. Then, for any integer n from 0 through N, the Taylor polynomial of order n generated by f at $x = a$ is the polynomial

$$P_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(k)}(a)}{k!}(x-a)^k + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

Note: We speak of a Taylor polynomial of order n rather than degree n because $f^{(n)}(a)$ may be zero.

Taylor Polynomials

Example: Find the Taylor series and the Taylor polynomials generated by $f(x) = e^x$ at $x = 0$.

Taylor Polynomials

Example: Find the Taylor series and the Taylor polynomials generated by $f(x) = \cos x$ at $x = 0$.

Taylor Polynomials

Example: Expand $f(x) = \ln(1 + x)$ at $x = 0$.

Taylor Polynomials

Example: Compute the Taylor polynomials of order 7 generated by $f(x) = \sinh x$ at $x = 0$.

Taylor Polynomials

Consider

$$f(x) = \begin{cases} 0, & x = 0 \\ e^{-1/x^2}, & x \neq 0 \end{cases}$$